An Overview on Cybersecurity Challenges and Solutions in the Internet of Everything Ecosystem

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***Abstract*—The Internet of Eveything (IoE) represents a dynamic ecosystem of people, processes, data, and things, offering many opportunities but also significant cybersecurity challenges. This survey presents a comprehensive literature review on the state-of-the-art in cybersecurity within the IoE landscape. It critically examines the complex threats and vulnerabilities inherent in highly interconnected systems, ranging from traditional cybersecurity risks to novel challenges introduced by the integration of different technologies. Special emphasis is placed on the evaluation of both hybrid human-AI collaborative approaches and fully autonomous security solutions, assessing their efficacy in mitigating these risks. By reviewing existing research, this survey aims to highlight key insights, identify emerging trends, and pinpoint gaps in current methodologies, thereby laying the groundwork for future research and practical advancements in IoE cybersecurity.**

***Index Terms*—Internet of Everything, Cybersecurity, Human-AI Collaboration, Autonomous Security Solutions**

I. Introduction

The proliferation of the Internet of Everything (IoE) has radically transformed how societies, industries, and individuals interact with technology, creating hyper-connected ecosystems that integrate people, processes, data, and physical objects seamlessly. While the IoE offers unprecedented opportunities for innovation, efficiency, and enhanced user experiences, it simultaneously introduces complex cybersecurity challenges that extend far beyond conventional information security paradigms.

In recent years, the exponential growth in IoE deployments across diverse sectors, ranging from smart homes and healthcare systems to industrial automation and critical infrastructures, has significantly expanded the attack surface, creating sophisticated vulnerabilities exploited by increasingly advanced cyber threats. Traditional cybersecurity measures often fall short within the IoE context due to the heterogeneous nature of connected devices, resource constraints, real-time operational demands, and the dynamic interactions between human and non-human actors.

In this overview, we aim to critically analyze and synthesize state-of-the-art research addressing cybersecurity challenges specific to the IoE ecosystem. By thoroughly examining contemporary literature, we explore conventional vulnerabilities and novel threats arising from interconnected devices and data flows, focusing particularly on innovative approaches such as hybrid human-AI collaborative security solutions and fully autonomous cybersecurity frameworks. We also highlight real-world applications through illustrative case studies, showcasing how theoretical insights translate into practical deployments. Ultimately, this paper seeks to identify gaps in current methodologies, provide meaningful insights into emerging trends, and propose directions for future research, contributing to the ongoing discourse on establishing robust, resilient cybersecurity infrastructures capable of supporting the dynamic demands of the IoE landscape. Practically, the main contributions of this paper can be summarized as follows:

- We highlight cybersecurity challenges in the IoE ecosystem by providing a discussion about methods, such as encryption and authentication, and by analyzing the studies present in the literature;
- We list and discuss different security solutions and strategies, ranging from hybrid security solutions, based on human and AI collaborative settings, to fully autonomous ones;
- We critically discuss some case studies and applications that showcase both the potential benefits and the inherent cybersecurity challenges of environments such as IoE.

This paper is structured as follows: in Section II we present a brief overview of current state-of-the-art. In Section III we present an overview of current challenges in the IoE ecosystems, as addressed in current literature. In Section IV we discuss currrent solutions and strategies to address the challenges in cybersecurity for IoE. In Section V we present some case studies and applications of cybersecurity in the context of IoE. Finally, in Section VI we draw some conclusions and discuss future directions of this research field.

II. Literature Review

Recent literature on cybersecurity in the Internet of Everything (IoE) highlights both the complexity and the urgency of securing increasingly interconnected ecosystems. Various approaches have been developed to address cybersecurity threats unique to IoE, each reflecting distinct advantages and inherent limitations.

One significant body of research focuses on enhancing authentication and encryption techniques tailored specifically

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for resource-constrained IoE devices. For instance, [1] introduces the Network Efficient Hierarchical Authentication (NEHA) algorithm, which significantly reduces latency and resource consumption by employing device-specific information to dynamically generate keys. Similarly, [2] proposes an integrated cybersecurity intelligence model leveraging a Hidden Markov Model (HMM) for efficient identification of illegitimate devices, emphasizing the need for dynamic resource allocation mechanisms that incorporate cybersecurity directly into operational decision-making.

Another critical dimension addressed in recent studies is the need for revised network architectures suitable for IoE environments. In [3], the authors proposed an Integrated Internet Architecture and Protocol (IoE-IAP) framework designed to handle multipath routing, scalability, and latency. Their layered approach effectively accommodates diverse communication pathways necessary for expansive IoE networks, which conventional architectures fail to adequately support.

The integration of machine learning algorithms for real-time threat detection in IoE has also gained substantial attention. [4] leverages Random Forest classifiers integrated with node categorization to enhance cybersecurity in smart grid environments, improving accuracy and resilience against cyber threats. Similarly, the authors of [16] demonstrated the effectiveness of deep auto-encoders enhanced by capsule graph convolution networks optimized via the sparrow search algorithm for anomaly detection in 6G IoE scenarios.

Forensic approaches adapted to IoE scenarios also contribute significantly to cybersecurity efforts. The authors of [5] developed a self-contained emulator facilitating the forensic examination of IoE devices and networks, bridging gaps between theoretical cybersecurity models and practical incident response requirements. This approach provides valuable tools for training investigators and evaluating forensic techniques under realistic conditions.

Hybrid Human-AI collaborative security solutions represent another promising research direction, integrating human oversight with automated precision to manage complex cybersecurity scenarios. [6] highlights the efficacy of such collaborative approaches in fields such as marketing analytics, while [8] and [9] illustrates similar benefits within industrial and manufacturing contexts. These studies emphasize the enhanced adaptability and nuanced decision-making capabilities offered by human-AI collaborations, although challenges persist, particularly concerning ethical implications and operational scalability.

Fully autonomous cybersecurity solutions are explored by the authors of [11], who introduced adaptive frameworks utilizing reinforcement learning within 5G-enabled IoT ecosystems. Such autonomous systems provide real-time, scalable threat detection and response capabilities, further enhanced by decentralized blockchain-based architectures proposed by [12]. Despite their advantages in speed and scalability, fully autonomous approaches continue to struggle with ethical dilemmas, adaptability to novel threats, and significant computational resource requirements.

While the existing literature effectively identifies and addresses diverse aspects of IoE cybersecurity, notable gaps remain. Authentication and encryption techniques, though optimized for efficiency, still face scalability challenges in highly heterogeneous device ecosystems. Similarly, revised network architectures show potential but often lack comprehensive validation in real-world environments. Machine learning models, although powerful for threat detection, frequently demand substantial computational resources not always feasible in IoE settings [19], [20].

Hybrid human-AI solutions balance human judgment with automated speed, yet they are hindered by ethical and scalability issues, especially under high operational demands or rapidly evolving threats. Conversely, fully autonomous solutions, while excellent in response times and scalability, suffer from a lack of nuanced ethical judgment and adaptability to unexpected threats [21], [22].

III. Cybersecurity Challenges in the IoE Ecosystem

Cybersecurity in the Internet of Everything (IoE) presents a unique and multifaceted challenge that extends beyond traditional IoT concerns by integrating devices, data, processes, and human interactions into a single interconnected framework. This convergence not only brings vulnerabilities once confined to isolated systems but also introduces new risks that require a more holistic approach. The wide variety of resource-constrained devices in IoE environments demands security solutions capable of protecting data integrity and confidentiality while operating efficiently across heterogeneous settings.

Conventional encryption and authentication methods can incur prohibitive computational overhead when applied to IoE scenarios. For instance, the Network Efficient Hierarchical Authentication (NEHA) algorithm leverages device-specific information, such as MAC addresses, to derive dynamic keys and segments data into optimized blocks, reducing latency and resource consumption compared to traditional methods like AES or Triple-DES [1]. Simultaneously, the dynamic nature of IoE necessitates resource allocation models that integrate cybersecurity intelligence into operational decision-making. One model employs a two-state Hidden Markov Model (HMM) to detect illegitimate devices efficiently, while robust encryption and decryption protocols secure task communications, transforming the scheduler into an intelligent resource allocator that enhances system performance and fortifies network integrity [2].

In light of these challenges, traditional network architectures require re-examination. An Integrated Internet Architecture and Protocol (IoE-IAP) framework redefines conventional protocols to better support the expansive and heterogeneous demands of IoE. This layered architecture model, which spans from perception and data-link layers to network and application tiers, addresses issues related to multipath routing, scalability, and latency by enabling more flexible and adaptive communication pathways [3].

TABLE I
A DETAILED DEPICTION OF SOME CYBERSECURITY CHALLENGES IN THE IOE ECOSYSTEM

Challenge	Description	Reference(s)	Underlying Cause(s)
Scalability of Authentication Mechanisms	Traditional authentication methods are inefficient for diverse, large-scale IoE deployments.	[1]	Heterogeneous devices, resource constraints, latency
Efficient Resource Allocation	Security solutions must integrate with dynamic operational demands in real-time systems.	[2]	Need for intelligent, adaptive security-aware schedulers
Outdated Network Architectures	Legacy architectures struggle with multipath routing, scalability, and latency.	[3]	Inflexibility of conventional layered architectures
Intrusion Detection in Smart Grids	Detecting sophisticated attacks requires machine learning and node categorization.	[4]	Critical infrastructure, large attack surface
Forensic Investigations in Dynamic Environments	Need for practical tools to replicate real-world scenarios for post-incident analysis.	[5]	Device diversity, lack of standardized forensic methods
Real-Time Operation Constraints	Security techniques must operate efficiently under low-latency, high-frequency conditions.	Implicit in multiple	Time sensitivity, limited processing power
Heterogeneity of Devices and Protocols	Vast diversity in hardware/software lead to incompatibility and varying vulnerabilities.	General Challenge	Fragmentation of IoE ecosystem
Ethical and Contextual Considerations in Automation	Automated systems may lack contextual or ethical understanding when handling threats.	[7], [21], [22]	Lack of human-in-the-loop or ethical modeling

The critical nature of many IoE applications, such as smart grids within the Internet of Energy, further complicates cybersecurity. Wireless Sensor Networks (WSNs) serving as the backbone for real-time monitoring and control in smart grids benefit from integrated traffic analysis and node categorization, combined with machine learning algorithms like Random Forest. This approach enhances the detection accuracy of cyber threats by analyzing network traffic patterns and classifying nodes based on sensitivity and workload, ensuring the reliability and resilience of power distribution systems [4]. Furthermore, practical challenges in forensic investigations within dynamic IoE environments are also addressed through the development of a self-contained emulator for IoE forensic examinations. This emulator replicates both the static firmware and dynamic network behaviors of IoE devices, providing a controlled testbed for evaluating forensic techniques, training investigators, and conducting in-depth analyses of cyber incidents. It effectively bridges the gap between theoretical models and real-world forensic requirements by enabling access to and analysis of data from a wide array of IoE devices and protocols [5].

Collectively, these studies illustrate that the cybersecurity challenges of IoE are deeply intertwined with its inherent heterogeneity, dynamic resource allocation, and the need for efficient, scalable authentication mechanisms, all compounded by the complexities of modern network architectures. The convergence of IoT devices with broader social and process elements demands a paradigm shift from conventional security protocols to integrated, intelligence-driven solutions. A schematic survey of the discussed challenges is also depicted in Table I.

IV. SECURITY SOLUTIONS AND STRATEGIES

As IoE ecosystems become increasingly complex, developing effective security solutions is critical to protect interconnected systems from sophisticated cyber threats. A wide range of approaches has emerged to address these challenges, broadly categorized into hybrid (Human-AI collaborative) and fully autonomous security solutions. Hybrid solutions leverage the complementary strengths of human expertise and AI-driven analytics, ensuring that all the decision-making and adaptive responses are integrated into the cybersecurity workflow. In contrast, fully autonomous solutions rely solely on advanced algorithms and machine learning models to detect, respond to, and mitigate threats in real time, reducing the need for human intervention. The following subsections detail these two strategies, exploring their architectures, advantages, and inherent challenges.

A. Hybrid (Human-AI Collaborative) Security Solutions

Hybrid (Human-AI Collaborative) security solutions for the IoE harness the complementary strengths of advanced automated systems and human expertise to tackle an increasingly complex cyber threat landscape. AI-driven analytics and machine learning algorithms continuously monitor, detect, and even predict emerging threats by processing large volumes of heterogeneous data. However, these systems are not left to operate autonomously; human oversight is integrated to interpret nuanced situations, adapt strategies in real time, and apply ethical considerations that pure automation might overlook.

Studies in marketing analytics demonstrate that combining AI's speed and precision with human judgment can optimize decision-making and personalize interventions more effectively than relying on either approach alone [6]. This principle translates well to cybersecurity, where AI quickly flags potential anomalies or cyber incidents in IoE networks, and human analysts then verify alerts, assess contextual risk factors, and determine appropriate remediation measures.

Research on autonomous systems, particularly within the framework of the Internet of Artificial Intelligence (IoAI), reveals that fully autonomous, human-disconnected systems, while efficient, often struggle with unforeseen situations and ethical dilemmas [7]. Such findings underscore the importance of retaining a human-in-the-loop approach, ensuring that experts can intervene when algorithmic outputs are ambiguous

TABLE II
A SUMMARY OF RECENT CYBERSECURITY SOLUTIONS IN IoE ENVIRONMENTS

Ref	Type	Core Technique / Approach	Application Context	Strengths	Weaknesses
[6]	Hybrid (Human-AI)	Human–AI collaboration in analytics	Marketing / Cybersecurity	Combines AI speed with human insight	Scalability and operational burden
[7]	Hybrid (Human-AI)	Internet of Artificial Intelligence (IoAI)	Autonomous systems	Highlights autonomy challenges, justifies human-in-loop	Ethical dilemmas and context-awareness issues
[8]	Hybrid (Human-AI)	Human–robot collaboration, ML	Industrial / Manufacturing	Human feedback enhances decision-making	Learning requires ongoing supervision
[9]	Hybrid (Human-AI)	Digital twins, collaborative robotics, augmentation	Industry 5.0 / Smart manufacturing	Improved safety, performance, and adaptability	Complexity in integration
[10]	Hybrid (Human-AI)	Self-learning humanoid robots with human oversight	Robotics / AI security	Accurate interpretation with ethical alignment	High system complexity
[11]	Fully Autonomous	Reinforcement learning-based adaptive framework	5G-enabled IoT	Real-time response, scalable	Lacks human ethical judgment
[12]	Fully Autonomous	Blockchain-based decentralized security strategy	IoE General	Data integrity, eliminates single points of failure	Resource and computational overhead
[13]	Fully Autonomous	Self-managed security cells	IoE General	Continuous, autonomous threat mitigation	Limited flexibility in novel or unseen scenarios

or when complex trade-offs require deeper contextual understanding.

Further evidence of the benefits of human-AI collaboration emerges from investigations into human–robot collaboration and machine learning in industrial settings [8]. These studies show that human feedback significantly enhances adaptive learning and decision-making capabilities. In cybersecurity for IoE, this collaborative model supports a layered approach: AI systems perform high-speed threat detection, pattern recognition, and data correlation across diverse network segments, while human experts use their experience and intuition to interpret insights, validate critical alerts, and refine defense strategies.

Research exploring collaborative robotics and digital twins within Industry 5.0 contexts also illustrates that integrating AI-driven automation with human decision-making enhances performance while ensuring safety and accountability. Digital twins and augmentation technologies not only improve production processes but also enable human operators to intervene when necessary, providing a model for human–AI collaboration that is directly applicable to cybersecurity in IoE environments [9].

Insights from work on delivering smarter, self-learning, autonomous humanoid robots further reinforce this approach. Advanced machine learning and cognitive models empower robotic systems to achieve rapid threat detection and adaptive responses. Yet, when these techniques are complemented by human expertise, the system's outputs are interpreted accurately, and countermeasures are implemented in alignment with broader strategic and ethical frameworks [10].

Overall, hybrid security solutions improve both the accuracy and speed of threat detection and enhance strategic response by blending algorithmic precision with human ethical oversight and situational awareness. This balanced synergy is crucial in the IoE ecosystem, where the integration of diverse devices, data flows, and human processes creates a dynamic environment that demands rapid, automated responses as well as thoughtful, context-aware decision-making.

B. Fully Autonomous Security Solutions

Fully autonomous security solutions for the IoE are engineered to operate with minimal human intervention by leveraging advanced AI, machine learning, and closed-loop feedback mechanisms to continuously monitor, detect, and respond to cyber threats in real time. These systems analyze data streams from a diverse array of IoE devices automatically, dynamically adapting their defense strategies as threat landscapes evolve. For instance, autonomous adaptive security frameworks developed for 5G-enabled IoT environments utilize reinforcement learning techniques that interact with the network environment, adjusting parameters on the fly to identify emerging threats, optimize resource allocation, and trigger countermeasures without manual oversight [11].

In addition to adaptive learning, fully autonomous solutions increasingly integrate decentralized architectures such as blockchain-based protocols to secure data exchanges and provide tamper-evident logging of security events. By decentralizing security operations, these systems eliminate single points of failure and enhance data integrity and transparency, which is particularly crucial in the heterogeneous and dynamic IoE ecosystem [11], [12]. This combination of decentralized data protection with autonomous, adaptive algorithms creates a resilient security infrastructure capable of managing the complex interactions and vulnerabilities inherent in IoE networks.

Further advancements focus on self-managed security cells that autonomously coordinate threat detection and response across multiple layers of an IoE network. These cells continuously assess risks, implement proactive defense measures, and reconfigure security protocols in response to evolving attacks. Such autonomous frameworks are essential for environments where IoE devices operate under varying conditions and require rapid, real-time decision-making to mitigate threats before they escalate [13].

Overall, fully autonomous security solutions deliver high-speed, data-driven responses through continuous learning and adaptive countermeasures. While they significantly reduce response times and operational overhead, the inherent trade-offs, such as the potential need for human intervention in unforeseen or ethically complex scenarios, underline the importance of further research into balancing full autonomy with occasional expert oversight. A bird's-eye view of the challenges addressed is also provided in Table II.

V. Case Studies and Applications

The proliferation of the Internet of Everything (IoE) has spurred a wide array of case studies and applications that showcase both the potential benefits and the inherent cybersecurity challenges of highly interconnected environments. Practical deployments span various sectors, from industrial control systems and smart manufacturing to healthcare and automotive, each illustrating unique vulnerabilities and the tailored measures taken to secure them.

One study examines how the integration of machine learning with traditional cybersecurity measures can enhance threat detection in industrial IoT settings. In this application, an anomaly-based intrusion detection system is implemented on an operational testbed, where machine learning algorithms continuously analyze network traffic to identify unusual patterns indicative of cyberattacks. The system effectively distinguishes between benign anomalies and malicious activity, enabling rapid and automated mitigation strategies without compromising production efficiency [14].

In another case, the deployment of a self-contained forensic emulator for IoE scenarios has provided researchers and practitioners with a controlled environment to simulate both static firmware behavior and dynamic network interactions. This emulator has been instrumental in training forensic investigators and evaluating novel forensic techniques, ensuring that emerging cyber threats are thoroughly analyzed and countered in real-world IoE deployments [15].

Another compelling case study focuses on anomaly detection in 6G IoE environments using advanced deep learning techniques. In this work, a dual deep auto-encoder is enhanced with capsule graph convolution and optimized via a sparrow search algorithm. By incorporating a novel loss function, the model generates a pseudo-abnormal bottleneck feature during training, effectively simulating abnormal data. This enables the decoder to learn to reconstruct normal data accurately while producing higher reconstruction errors for abnormal inputs. As a result, the approach significantly improves detection performance even when abnormal data closely resemble normal data [16].

Another case study illustrates the challenges and innovative solutions in IoT-based healthcare applications. This work begins by defining IoT as a network of physical objects—such as gadgets, vehicles, and buildings, embedded with sensors, software, and connectivity, enabling autonomous data collection and communication. It emphasizes how the exponential growth of heterogeneous devices necessitates robust, interoperable networking protocols to ensure seamless interconnection. Focusing on smart healthcare, the study shows how IoT-based systems can monitor, record, and analyze patient data through technologies like wireless body area networks, implants, and wearable sensors, thereby improving service quality and reducing costs. Notably, the framework compares the performance of two application layer protocols, CoAP and MQTT, and introduces an adaptive switching-based data communication model that optimizes network efficiency and availability in response to conditions such as large payloads and network congestion. This adaptive approach significantly enhances the reliability and effectiveness of smart healthcare systems, contributing to improved quality of life and economic growth in IoT-driven environments [17].

Finally, another study addresses the cybersecurity challenges of RPL-based networks within the IoE by focusing on the need to secure routing and information sharing in heterogeneous, resource-constrained environments. The work underscores that while IoT connects various devices, from sensors and cameras to vehicles and industrial systems, there are significant vulnerabilities, especially in RPL protocols that underpin low-power and lossy networks. Routing attacks exploiting node rank and version number can lead to congestion, increased latency, and energy inefficiencies. To counter these challenges, the study develops a collaborative intrusion detection system (CIDS) that dynamically controls traffic congestion using workload and queue information, integrates a fuzzy logic-based trust evaluation mechanism to accurately assess node reliability, and employs optimization techniques such as genetic algorithms and Q-learning to efficiently detect coordinated attacks. Simulation results, obtained using the Cooja simulator on Contiki-3.x, demonstrate that this CIDS achieves superior detection accuracy while reducing network delay and energy consumption, making it a robust solution for high-density IoE applications like smart cities and healthcare [18].

These case studies collectively illustrate how addressing cybersecurity challenges within diverse IoE applications requires tailored solutions that integrate advanced analytics, adaptive strategies, and interdisciplinary collaboration. They highlight both successful practical deployments and remaining complexities that inform future research and development directions in IoE security.

VI. Conclusion and Future Directions

In this paper, we conducted an overview of the cybersecurity landscape in IoE environments. Such environments are characterized by inherent complexity, driven by diverse interconnected devices and dynamic interactions between humans and automated systems, and therefore the urge of studying them from the point of view of cybersecurity arises. Current approaches, including hybrid Human-Artificial Intelligence collaborations and fully autonomous solutions, offer promising avenues to mitigate security threats effectively. Nevertheless, each approach has limitations, particularly concerning ethical considerations, adaptability to unforeseen scenarios, and resource efficiency in highly heterogeneous environments. In this survey, we have contributed to this field in different ways. We first identified and analyzed the fundamental cybersecurity challenges specific to the IoE landscape, highlighting different issues such as scalability, real-time constraints and architectural limitations. Then, we provided a structured and comprehensive overview of existing security strategies, outlining their respective strengths and limitations. Also, we mapped these strategies to relevant literature and use cases especially, thus aiming to offer a consolidated perspective on how different techniques address specific threats.

In our opinion, future research should prioritize enhancing adaptive cybersecurity mechanisms that can dynamically respond to evolving threats in real-time. This includes developing lightweight, scalable encryption methods suitable for resource-constrained devices, improving anomaly detection algorithms through advanced machine learning techniques, and integrating ethical guidelines explicitly into autonomous decision-making frameworks. Moreover, interdisciplinary collaboration between cybersecurity experts, ethicists, and industry stakeholders will be crucial to navigate the complex ethical dilemmas inherent in autonomous security operations. Ultimately, addressing these challenges will require continuous adaptation and rigorous validation through practical deployments, ensuring that the IoE ecosystem remains secure, trustworthy, and resilient against emerging cyber threats.

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